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ScienceDirect
Annals of Nuclear Energy 34 (2007) 406–416
annals of NUCLEAR ENERGY
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Application of cellular neural network (CNN) method to the nuclear reactor dynamics equations

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Received 19 September 2006; received in revised form 15 January 2007; accepted 9 February 2007
Available online 2 April 2007

Abstract

This paper describes the application of a multilayer cellular neural network (CNN) to model and solve the nuclear reactor dynamic equations. An equivalent electrical circuit is analyzed and the governing equations of a bare, homogeneous reactor core are modeled via CNN. The validity of the CNN result is compared with numerical solution of the system of nonlinear governing partial differential equations (PDE) using MATLAB. Steady state as well as transient simulations, show very good comparison between the two methods. We used our CNN model to simulate space-time response of different reactivity excursions in a typical nuclear reactor.

On line solution of reactor dynamic equations is used as an aid to reactor operation decision making. The complete algorithm could also be implemented using very large scale integrated circuit (VLSI) circuitry. The efficiency of the calculation method makes it useful for small size nuclear reactors such as the ones used in space missions.

1. Introduction

An understanding of time dependent behavior of neutron population in a nuclear reactor in response to a planned change in reactor conditions or unplanned and abnormal conditions is of utmost importance to the safe and reliable operation of nuclear reactors. The dynamics of a nuclear reactor involves the coupling of space-time neutron kinetics with the thermal-hydraulics feedbacks. For neutronics alone (also known as kinetics), a tremendous amount of physical modeling and mathematical computation have been carried out.

In recent years, solutions have been put forward to the more sophisticated problems with time varying coefficients. In most of these solutions, there are some approximations generally involved or applied effectively only to certain types of problems. Among the methods that have been used in the past or are currently in use are numerical integration using Simpson's rule (Keepin and Cox, 1960), finite difference methods (Brown, 1957), Runge–Kutta procedures (Allerd and Carter, 1958; Blue and Hoffman, 1963; Sanchez, 1989) and fully analytic approach for reactivity oscillations analysis (Ravetto, 1997). Other methods are based on integral equation formulation, where the slowly varying factor in each integrand represented by an assumed functional form (Kaganove, 1960; Adelr, 1961; Hansen, 1965; Porsching, 1966). These methods suffer from one or more disadvantages (Vigil, 1967). Recently software packages such as MATLAB (Hadad, 2004) and Maple (Faghihi and Hadad, 2004) have been used to solve nonlinear system of PDEs.

Cellular neural network (CNN) has been used to solve such problems (Chua and Roska, 1993, 2002; Chua et al., 1995). Though originally designed to solve the PDEs governing the electrical circuits, CNN has successfully been applied to non-electrical problems as well. Parallel data processing in CNN reduces the processing time and makes it possible to solve a real 3D, time dependent models of nuclear reactors in real time. Thus, the CNN implementation avoids the exponential increase in complexity as the

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doi:10.1016/j.anucene.2007.02.001

scale and dimension of the reactor core nodes increase. CNN has been applied to the simulation of several physical systems governed by PDEs (Kozek et al., 1995; Balsi et al., 1995). Such networks are discrete space, continuous-time, continuous-state systems that easily lend themselves to such simulations by applying an approximation in space domain using finite differences.

In this study, reactor dynamic equations are modeled and solved using a CNN paradigm. Our network is a multi-layer cellular network which enables us to study steady state and transient conditions of neutron flux density due to a reactivity excursion of a nuclear reactor.

1.1. Architecture of CNN

The basic unit of a CNN is called a cell. It contains linear and nonlinear circuit elements which typically are capacitors, resistors, controlled sources and an independent source. The structure of a two-dimensional (2-D) 3×3 CNN is shown in Fig. 1.

Consider an $M \times N$ cellular neural network, having M rows and N columns. We call the cell on the ith row and jth column, cell (i, j) and denote it by $C(i, j)$.

A typical example of a cell is shown in Fig. 2, where the suffixes x and y denote the state and output, respectively. The node voltage υ_{xij} of $C(i, j)$ is defined as the state of the cell whose initial condition is assumed to have a magnitude less than or equal to 1. The node voltage is called the output of $C(i, j)$. Observe from Fig. 2 each cell contains one independent current source, one linear-capacitor C, two linear resistors R_x and R_y and linear voltage-controlled current sources, which are coupled to its neighboring cells via the feedback from the output voltage $\upsilon_{yi'j'}$ of each neighbor cell $C(i',j')$.

In particular, $I_{xy}(i, j{:}i', j')$ are linear voltage-controlled current sources with the characteristics $I_{xy}(i,j:i',j') = A(i,j:i',j')\upsilon_{yi'j'} \cdot A(i,j:i',j')$ is the output feedback from cell (i', j') to cell (i, j). The only nonlinear element in each cell is a piecewise-linear voltage-controlled current source, $I_{xy} = (1/R_y)f(\upsilon_{xij})$, with characteristic $f(\cdot)$ as shown in Fig. 3.

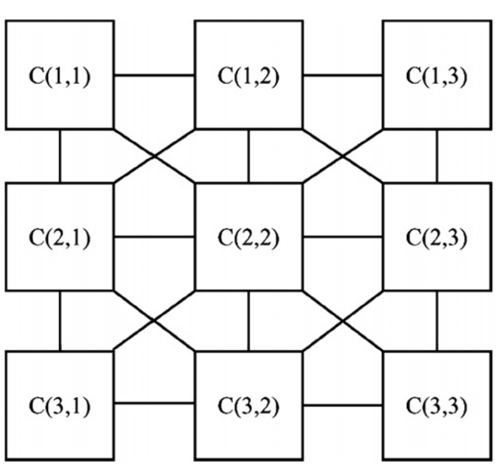

Fig. 1. A 2-D cellular neural network of size 3×3. Squares are the circuit units called cells. Links between the cells indicate interactions between them.

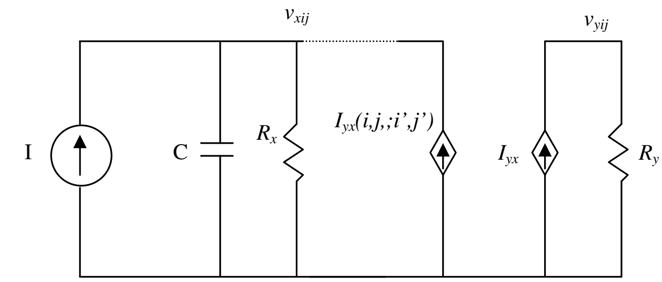

Fig. 2. A cell circuit where C is a linear-capacitor; R_x and R_y are resistors; I is an independent current source ;$I_{yx}(i; j; i ; j)$ is linear voltage-controlled current source, I_{yx} is a piecewise-linear voltage-controlled current source with its characteristic $f(.)$ as shown in Fig. 3.

The equivalent block diagram of a CNN cell is shown in Fig. 4. The first-order nonlinear differential equation defining the dynamics of a CNN can be derived as follows:

State equations:

$$C\frac{d\upsilon_{xij}(t)}{dt} = -\frac{1}{R}\upsilon_{xij}(t) + \sum_{C(k,l)\in N_r(i,j)} A(i,j;k,l)\upsilon_{ykl}(t) + \sum_{C(k,l)\in N_r(i,j)} B(i,j;k,l)\upsilon_{ukl} + I \tag{1}$$

$$1 \leqslant i \leqslant M;\quad 1 \leqslant j \leqslant N.$$

Output equation:

$$\upsilon_{yij}(t) = \frac{1}{2}(|\upsilon_{xij}(t) + 1| - |\upsilon_{xij}(t) - 1|). \tag{2}$$

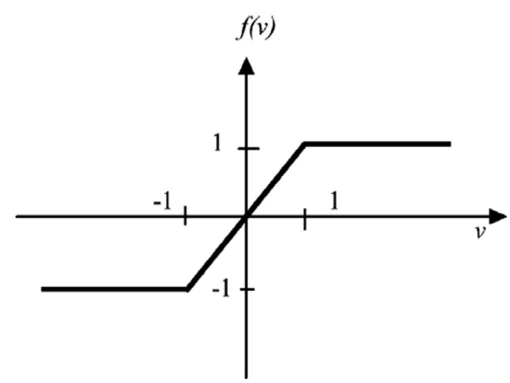

Fig. 3. The characteristic of the nonlinear controlled source.

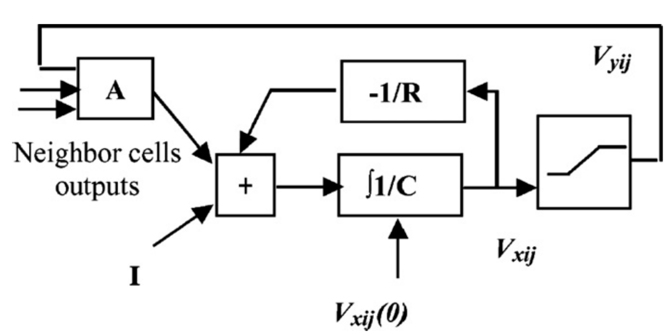

Fig. 4. Equivalent block diagram of a CNN cell.

Input equation:

$$v_{uij} = E_{ij} \quad 1 \leqslant i \leqslant M;\ 1 \leqslant j \leqslant N. \tag{3}$$

Constraint conditions:

$$|v_{xij}(0)| \leqslant 1 \quad 1 \leqslant i \leqslant M;\ 1 \leqslant j \leqslant N, \tag{4}$$

$$|v_{uij}| \leqslant 1 \quad 1 \leqslant i \leqslant M;\ 1 \leqslant j \leqslant N,$$

$$A(i,j;k,l) = A(k,l;i,j) \quad 1 \leqslant i,k \leqslant M;\ 1 \leqslant j,l \leqslant N,$$
$$C > 0, \quad R_x > 0. \tag{5}$$

1.2. Solving PDEs using CNN

When solving PDEs, four variables are represented in either continuous or discrete form: time, value (of the state variables), interaction (parameters), and space. When numerical computing applied to nonanalytic solution of PDEs, some kind of spatial discretization is always performed. This procedure transforms a PDE into a set of ordinary differential equations (ODEs). This means that, the original spatially continuous system is transformed into an array of small, discrete, interacting system. The CNN paradigm is a natural and flexible framework for describing locally interconnected, simple, dynamical systems that have an array-like structure (Roska et al., 1995).

CNNs have been successfully applied to problems ranging from heat transfer (Krstic, 2001) to nonlinear wave (Kozek et al., 1995; Civalleri and Gilli, 1995) and image processing (Crounse and Chua, 1995). In these applications spatial discretization has been applied. The PDEs are transformed into a system of ODEs which are identified as the state equation of a CNN. The discretization in space is made in equidistant discrete steps in both directions.

2. Simulation of the nuclear reactor core using CNN

Reactor core dynamics are represented by linear and nonlinear differential equations with varying coefficients, which are functions of the core operating conditions (power level, coolant and fuel temperatures, coolant density, poison buildup, burn-up rate, etc.) (Stacy, 2001). The space-time neutron flux is obtained by solving a minimum of six sets of PDEs consisting:

(a) The neutron diffusion equation.
(b) The delayed neutron precursor equation(s).
(c) The poison built up and decay equations.
(d) The decay heat equations.
(e) Energy balance equations.
(f) Mass balance equations.

By using a suitable nodalization of the reactor core, finite difference and finite element methods have been used to solve the above equations numerically. The accuracy of the calculations is determined by core nodalization and neutron physical equations and constants. Neutron cross sections, for example, depend on fuel and moderator temperatures, moderator density, and neutron poisons concentration, necessitating suitable interpolation. Hence, finding a precise time-space flux distribution in the core necessitates solving a large number of equations simultaneously which in turn, require a large amount of time and memory on digital computers.

In our study we used a simplified dynamic model consisting of:

- Two group diffusion.
- One equivalent group of neutron precursor density.
- Moderator feedback coefficient.
- Neutron poisons consisting of xenon and iodine.
- No decay heat.
- Prompt and delayed neutron are born in the fast groups.

2.1. Reactor dynamic model

The dynamic model differs from kinetic model by including the reactivity feedbacks from fuel and moderator.

2.1.1. Time dependent 2-group diffusion equation

$$\begin{cases} \frac{1}{v^{(1)}}\frac{\partial\phi^{(1)}}{\partial(t)} = D^{(1)}\nabla^2\phi^{(1)} - \Sigma_{\mathrm{r}}^{(1)}\phi^{(1)} + \frac{1}{k}(1-\beta) \\ \qquad\qquad \times \sum_{g=1}^{2} v\Sigma_{\mathrm{f}}^{(g)}\phi^{(g)} + \lambda C, \\ \frac{1}{v^{(2)}}\frac{\partial\phi^{(2)}}{\partial t} = D^{(2)}\nabla^2\phi^{(2)} - \Sigma_{\mathrm{r}}^{(2)}\phi^{(2)} + \Sigma_{\mathrm{s}}^{(12)}\phi^{(1)}, \end{cases} \tag{6}$$

where g is the energy group, $v^{(g)}$ neutron velocity of g group in (cm/s), $\phi^{(g)}$ neutron flux of g group in (neutron/cm^2 s), and $\Sigma_{\mathrm{r}}^{(g)}$ removal and $\Sigma_{\mathrm{f}}^{(g)}$ fission macroscopic cross sections of group g in (cm^{-1}), respectively, $\Sigma_{\mathrm{s}}^{(12)}$ macroscopic scattering cross section from energy group 1 to 2 in (cm^{-1}), $D^{(g)}$ neutron diffusion coefficient of group in (cm^{-1}), β the fraction of delayed neutrons, v average neutrons produced per fission, λ decay constant of precursors in (s^{-1}), and C is the one group delayed neutron precursors concentration in (nuclei/cm^3).

2.1.2. Delayed neutron precursor equation

The effective one group delayed neutron precursor density is (Stacy, 2001):

$$\frac{\partial C}{\partial t} = -\lambda C + \beta\sum_{g=1}^{2} v\Sigma_{\mathrm{f}}^{(g)}\phi^{(g)}. \tag{7}$$

2.1.3. Poison equations

Xenon and iodine have the highest yield and absorption cross sections among other poisons, then their concentrations influence neutron population more effectively than other poisons.

$$\begin{cases} \frac{\partial \mathrm{Xe}}{\partial t} = (-\lambda_{\mathrm{Xe}} - \sigma_a^{\mathrm{Xe}}\phi^{(2)})\mathrm{Xe} + \lambda_{\mathrm{I}}\mathrm{I} + \gamma_{\mathrm{Xe}}\sum_{g=1}^{2}\Sigma_{\mathrm{f}}^{(g)}\phi^{(g)}, \\ \frac{\partial \mathrm{I}}{\partial t} = -\lambda_{\mathrm{I}}\mathrm{I} + \gamma_{\mathrm{I}}\sum_{g=1}^{2}\Sigma_{\mathrm{f}}^{(g)}\phi^{(g)}, \end{cases} \tag{8}$$

where Xe and I are xenon and iodine concentrations in (nuclei/cm^3), λ and γ are decay and yield constants, and σ_a^{Xe} is the thermal microscopic absorption cross section for xenon. The macroscopic removal cross section in Eq. (8) is affected by Xe and I concentrations.

2.1.4. Temperature feedback equation

Temperature feedback affects the thermal neutron flux and reactivity through the following equations:

$$\begin{cases} \frac{\mathrm{d}T}{\mathrm{d}t} = A_k\phi^{(2)} - \kappa T(t), \\ \rho(t) = \rho_{\mathrm{ext.}}(t) - \alpha_T T(t), \end{cases} \tag{9}$$

where T is the coolant and moderator temperature, ρ(t) is the instantaneous reactivity made up of the external reactivity $\rho_{ext}(t)$ and reactivity feedbacks $\alpha_{\mathrm{T}}T$.

2.2. CNN model for reactor dynamic equations

A multilayer CNN network nodalization, with symmetric nodalization in a discretized Cartesian coordinates is applied. At each grid point, the neutron flux is $\phi^{(g)}(x, y, z)$ and is associated with $\phi^{(g)}$ $(i\Delta x, j\Delta y, k\Delta z)$. Using Taylor series expansion in space, the diffusion equation becomes:

$$\begin{cases} \frac{1}{\vartheta^{(1)}}\frac{\partial\phi^{(1)}_{i,j,k}}{\partial t} = \frac{-1}{\frac{1}{(\Sigma_{\mathrm{r}}^{(1)}-(1-\beta)v\Sigma_{\mathrm{f}}^{(1)})}}\phi^{(1)}_{i,j,k} + \lambda \cdot C_{i,j,k} + (1-\beta)v\Sigma_{\mathrm{f}}^{(2)}\phi^{(2)}_{i,j,k} \\ \qquad -2D^{(1)}\left[\frac{1}{\Delta x^2}+\frac{1}{\Delta y^2}+\frac{1}{\Delta z^2}\right]\phi^{(1)}_{i,j,k} \\ \qquad +\frac{D^{(1)}}{\Delta x^2}\left[\phi^{(1)}_{i-1,j,k}+\phi^{(1)}_{i+1,j,k}\right] \\ \qquad +\frac{D^{(1)}}{\Delta y^2}\left[\phi^{(1)}_{i,j-1,k}+\phi^{(1)}_{i,j+1,k}\right] \\ \qquad +\frac{D^{(1)}}{\Delta z^2}\left[\phi^{(1)}_{i,j,k-1}+\phi^{(1)}_{i,j,k+1}\right] \end{cases} \tag{10}$$

$$\begin{cases} \frac{1}{\vartheta^{(1)}}\frac{\partial\phi^{(1)}_{i,j,k}}{\partial t} = \frac{-1}{\frac{1}{(\Sigma_{\mathrm{r}}^{(1)}-(1-\beta)\vartheta\Sigma_{\mathrm{f}}^{(1)})}}\phi^{(1)}_{i,j,k} + \lambda \cdot C_{i,j,k} + (1-\beta)\vartheta\Sigma_{\mathrm{f}}^{(2)}\phi^{(2)}_{i,j,k} \\ \qquad -2D^{(1)}\left[\frac{1}{\Delta x^2}+\frac{1}{\Delta y^2}+\frac{1}{\Delta z^2}\right]\phi^{(1)}_{i,j,k} \\ \qquad +\frac{D^{(1)}}{\Delta x^2}\left[\phi^{(1)}_{i-1,j,k}+\phi^{(1)}_{i+1,j,k}\right] \\ \qquad +\frac{D^{(1)}}{\Delta y^2}\left[\phi^{(1)}_{i,j-1,k}+\phi^{(1)}_{i,j+1,k}\right] \\ \qquad +\frac{D^{(1)}}{\Delta z^2}\left[\phi^{(1)}_{i,j,k-1}+\phi^{(1)}_{i,j,k+1}\right] \end{cases} \tag{11}$$

Comparison of Eqs. (10) and (11) with (1), results in 3-D, 2 layer CNN cells with following constants:

$$\begin{cases} C(\phi^{(g)}) = \frac{1}{\vartheta^{(g)}}, \quad g = 1,2, \\ R_x^{(\phi^{(2)})} = \frac{1}{\Sigma_{\mathrm{r}}^{(2)}}, \\ R_x^{(\phi^{(1)})} = \frac{1}{(\Sigma_{\mathrm{r}}^{(1)}-(1-\beta)v\Sigma_{\mathrm{f}}^{(1)})}, \\ A^{(\phi^{(g)})}(i,j,k,i,j,k) = -2D^{(g)}\left[\frac{1}{\Delta x^2}+\frac{1}{\Delta y^2}+\frac{1}{\Delta z^2}\right]. \end{cases} \tag{12}$$

$$\begin{cases} A^{(\phi^{(g)})}(i,j,k,i+1,j,k) = \frac{D^{(g)}}{\Delta x^2}, \\ A^{(\phi^{(g)})}(i,j,k,i,j,k+1) = \frac{D^{(g)}}{\Delta z^2}, \end{cases} \tag{13}$$

$$\begin{cases} A^{(\phi^{(1)}\phi^{(2)})}(i,j,k,i,j,k) = (1-\beta)v\Sigma_{\mathrm{f}}^{(2)}, \\ A^{(\phi_c^{(1)})}(i,j,k,i,j,k) = \lambda, \\ A^{(\phi^{(2)}\phi^{(1)})}(i,j,k,i,j,k) = \Sigma_{\mathrm{s}}^{(12)}, \end{cases} \tag{14}$$

where $C^{(n)}$ and $R_x^{(n)}$ are the nth layer capacitance and resistance, $A^{(nm)}(i,\ j,\ k,\ i',\ j',\ k')$ denotes linear voltage-controlled current source gain which interacts node i, j, k in nth layer with i', j', k' in mth layer of gth energy group.

Similarly, the equivalent CNN equations for delayed neutron precursors, poisons and temperature feed back are as follows:

$$\begin{cases} C^{(c)} = 1, \\ R_x^{(c)} = \frac{1}{\lambda}, \\ A^{(c\phi^{(g)})}(i,j,k,i,j,k) = \beta v\Sigma_{\mathrm{f}}^{g}, \quad g = 1,2, \end{cases} \tag{15}$$

$$\begin{cases} C^{(\mathrm{Xe})} = 1, \\ C^{(\mathrm{I})} = 1, \\ R_x^{(\mathrm{I})} = \frac{1}{\lambda_{\mathrm{I}}}, \\ R_x^{(\mathrm{Xe})}(\phi^{(2)}) = \frac{1}{\lambda_{\mathrm{Xe}}+\sigma_a^{\mathrm{Xe}}\phi^{(2)}_{i,j,k}}, \\ A^{(\mathrm{I}\phi^{(g)})}(i,j,k,i,j,k) = \gamma_{\mathrm{I}}\Sigma_{\mathrm{f}}^{(g)}, \\ A^{(\mathrm{Xe}\phi^{(g)})}(i,j,k,i,j,k) = \gamma_{\mathrm{Xe}}\Sigma_{\mathrm{f}}^{(g)}, \\ A^{(\mathrm{XeI})}(i,j,k,i,j,k) = \lambda_{\mathrm{I}}, \end{cases} \tag{16}$$

where $R_x^{(\mathrm{Xe})}(\phi^{(2)})$ is a voltage-controlled resistor (VCR) that is a function of neutron thermal flux $\phi^{(2)}_{i,j,k}$.

Iodine cross sections and xenon absorption cross section for fast neutrons are ignored. Therefore, to include poison effects, $R_x^{(\phi^{(2)})}$ in is replaced by a VCR as:

$$R_x^{(\phi^{(2)})}(\mathrm{Xe}_{i,j,k}) = \frac{1}{\Sigma_{\mathrm{r}}^{(2)} + \sigma_a^{\mathrm{Xe}} \cdot \mathrm{Xe}_{i,j,k}}, \tag{17}$$

Temperature feedback is modeled in CNN as:

$$R_x^T = -\frac{1}{\gamma}, \quad A^{(\phi^{(2)}T)} = A_k. \tag{18}$$

Eventually, a 6 layer, three dimensional CNN is needed to model the reactor dynamic equations. The model is expanded as the number of state variables (unknowns) is increased. Each layer of this network contains a discretized three dimensional space.

Table 1
Neutronic characteristics of a bare-homogeneous reactor (Hetrick, 1993; Duderstadt and Hamilton, 1976)

Energy group	Fast ($g = 1$)	Thermal ($g = 2$)
D (cm)	1.2627	0.3543
Σ_{r} (cm)$^{-1}$	0.02619	0.1210
$v\Sigma_{\mathrm{r}}$ (cm)^1	0.00847	0.18514
Σ_{s} (cm)^1	0.01412	0.0
V	1.25×10^7 (cm/s)	2.5×10^5 (cm/s)
$\gamma_1 = 6.38\%$	$\gamma_{\mathrm{Xe}} = 0.228\%$	$\alpha_{\mathrm{T}} = 0.016$
$\lambda_1 = 2.87 \times 10^{-5}$ (s)$^{-1}$	$\lambda_{\mathrm{Xe}} = 2.09 \times 10^{-5}$ (cm)^1	$\gamma = 0.0267$
$\lambda = 0.08$ (s)$^{-1}$	$\beta = 0.0065$	$A_{\mathrm{k}} = 12.5$

Table 2
Typical input file for the circuit simulator HSPICE for the reactor core

*Precursors equation, node no. 04 02 11							
******							
C10040211	10040211	0	1.0	ic=0.94642983492166 v			
R10040211	10040211	0	12.50000				
G100402111	0	10040211	21040211	0	0.005509		
G100402112	0	10040211	22040211	0	0.120341		
G20040211	0	20040211	10040211	0	1.0	MAX = +1	MIN = 0
R20040211	20040211	0	1.0				
*Neutron diffusion equation, energy group 1, node no. 04 02 11							
******							
R11040211	11040211	0	r = ‘0.01/(0.02629−(1 − A + ALF ∗ v(25040211)) ∗ 0.008414945)’				
R21040211	21040211	0	1.0				
C11040211	11040211	0	8u	ic=0.94642983492166 v			
G110402111	0	11040211	21050211	0	1.997864		
G110402112	0	11040211	21030211	0	1.997864		
G110402113	0	11040211	21040311	0	1.997864		
G110402114	0	11040211	21040111	0	1.997864		
G110402115	0	11040211	21040212	0	2.945830		
G110402116	0	11040211	21040210	0	2.945830		
G110402117	0	11040211	22040211	0	‘18.393659 ∗ (1 − A + ALF ∗ v(25040211))’		
G110402118	0	11040211	21040211	0	−13.883116		
G110402119	0	11040211	20040211	0	0.080000		
G21040211	0	21040211	11040211	0	1.0	MAX=+1	MIN=0
*Neutron diffusion equation, energy group 2, node no. 04 02 11							
******							
R12040211	12040211	0	r = ‘1/(0.121000 + 2.7 ∗ 1e − 2 ∗ v(24040211))’				
R22040211	22040211	0	1.0				
C12040211	12040211	0	4.0u	ic=0.94642983492166 v			
G120402111	0	12040211	22050211	0	0.005606		
G120402112	0	12040211	22030211	0	0.005606		
G120402113	0	12040211	22040311	0	0.005606		
G120402114	0	12040211	22040111	0	0.005606		
G120402115	0	12040211	22040212	0	0.008266		
G120402116	0	12040211	22040210	0	0.008266		
G120402117	0	12040211	21040211	0	0.014120		
G120402118	0	12040211	22040211	0	−0.038955		
G120402119	0	12040211	20040211	0	0.000000		
G22040211	0	22040211	12040211	0	1.0	MAX=+1	MIN=0
*Iodine equation, node no. 04 02 11							
******							
C13040211	13040211	0	10	ic=0.94642983492166 v			
R13040211	13040211	0	3484.320557				
G13040211	0	13040211	22040211	0	0.004876		
G130402111	0	13040211	21040211	0	0.000213		
G23040211	0	23040211	13040211	0	1.0	MAX=+1	MIN=0
R23040211	23040211	0	1.0				
*Xenon equation, node no. 04 02 11							
C14040211	14040211	0	10	ic=0.94642983492166 v			
R14040211	14040211	0	r = ‘0.1/(2.1e − 5 + 2.7e − 3 ∗ v(22040211))’				
G140402111	0	14040211	23040211	0	0.000290		
G140402112	0	14040211	22040211	0	0.000177		
G140402113	0	14040211	21040211	0	0.000008		
G24040211	0	24040211	14040211	0	1.0	MAX=+1	MIN=0
R24040211	24040211	0	1.0				
*Feedback equation, node no. 04 02 11							
C15040211	15040211	0	1	ic=0.94642983492166V			
R15040211	15040211	0	37.453184				
G150402113	0	15040211	22040211	0	4.1554363		
G25040211	0	25040211	15040211	0	1.0	MAX=+1	MIN=0
R25040211	25040211	0	1.0				

3. Results of case studies

Once the model programmed in HSpice and the boundary and initial conditions defined, the model was verified for satisfying the normal reactor operating conditions (in absence of any external perturbations). This objective was achieved via the selection of suitable reactivity feedback coefficients and neutron generation time (Duderstadt and Hamilton, 1976; Hetrick, 1993). In addition to reactivity coefficients, an excess reactivity insertion was needed to bring the system to steady sate. This is due to the presence of negative reactivity from poison produced in fission. Despite some difficulties found in reproduction of previous models (Boroushaki et al., 2005), the time varying flux in our model converged to steady state value without any non-physical parameter modification.

The dynamic model in a bare, homogenous, cubical reactor core of 111 cm sides is simulated by the CNN model as described in Section 2. Symmetrical grid in reactor core mapped onto a multilayer 3-D CNN. The neutronic data of the core are shown in Table 1. The reactor core is controlled by 4 control rods and the change of boric acid concentration. The Cartesian space is discretized into *10 × 10 × 18* nodes for a six layer CNN network which makes a total of *10 800* nodes. The black (non-reentrant) boundary condition is used for all case studies.

Since each of the state variables (thermal flux, fast flux, precursor concentration, xenon, and iodine concentrations and temperature) are varying in a different range, we normalized these parameters as follows:

$$\begin{cases} \phi_{\mathrm{N}}^{(1)} = \frac{\phi^{(1)}}{10^{15}} \ (\text{neutron/cm}^2\ \text{s}), \\ \phi_{\mathrm{N}}^{(2)} = \frac{\phi^{(2)}}{10^{15}} \ (\text{neutron/cm}^2\ \text{s}), \\ C_{\mathrm{N}} = \frac{C}{10^{13}} \ (\text{nuclei/cm}^3), \\ I_{\mathrm{N}} = \frac{I}{10^{16}} \ (\text{nuclei/cm}^3), \\ \mathrm{Xe}_{\mathrm{N}} = \frac{\mathrm{Xe}}{10^{16}} \ (\text{nuclei/cm}^3), \\ T_{\mathrm{N}} = \frac{T}{100} \ (^{\circ}\text{C}), \end{cases} \tag{19}$$

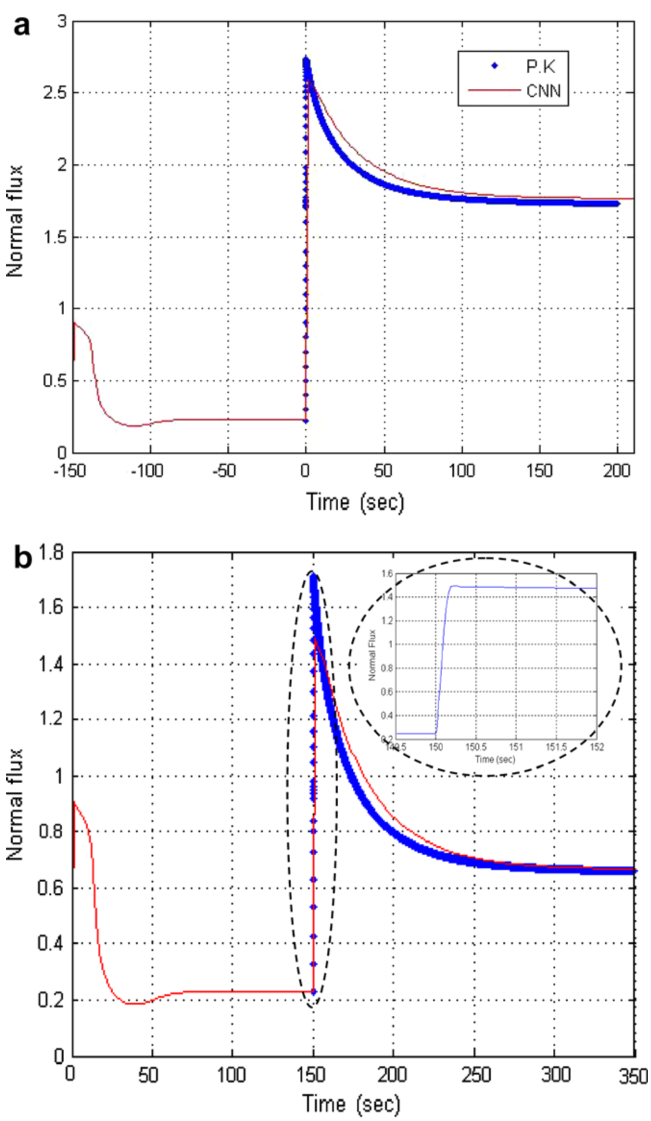

Fig. 5. The results of PK and CNN model for space independent neutron flux variations (a) + 0.5\$ *and* (b) + 0.1\$ positive reactivity perturbations.

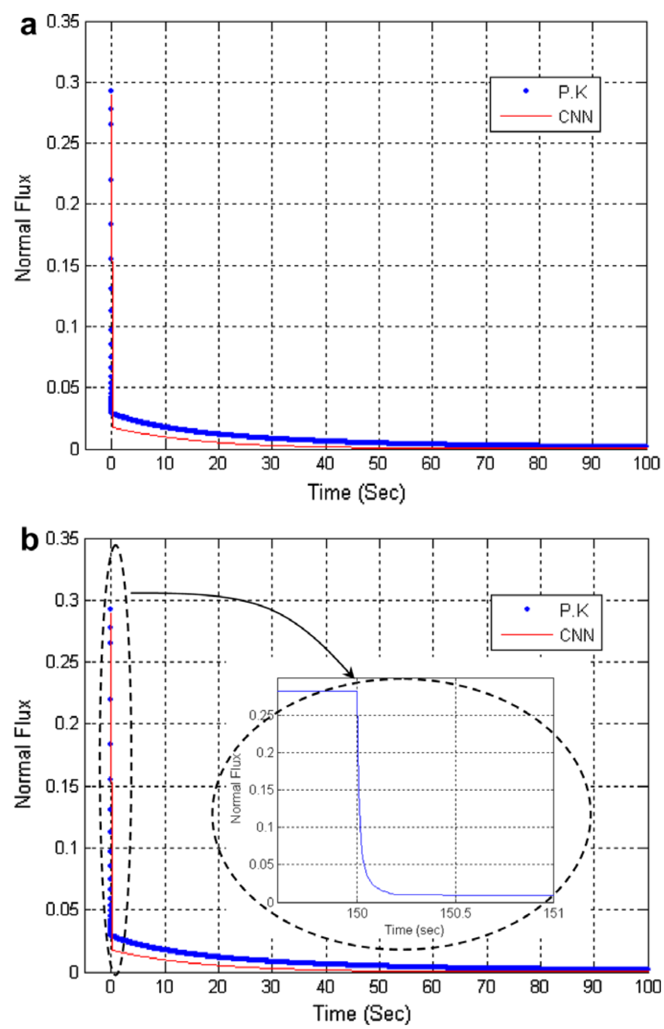

Fig. 6. The results of PK and CNN model for space independent neutron flux variations (a) −0.5\$ and (b) −0.1\$ reactivity perturbations.

where the subscript N stands for a normalized variable. Using the reactor core data in Table 1 and normalization used in Eq. (19), values of the CNN elements are then calculated. CNNs are nonlinear dynamical systems; there are presently few analytical methods for studying their transient behavior. Implementation of CNN model is carried out using HSpice software Package which is a powerful version of PSPICE software package. HSpice has been successfully applied for complicated circuitry modeling (Kim et al., 2005). Table 2 shows a segment of HSpice input code.

It is remarkable that the only approximation committed in CNN modeling is the space discretization of the Laplacian operator in the diffusion equation. Space discretization, however, should be chosen carefully in the neighborhood of fast varying flux (or cross section) regions in inhomogeneous reactors (Duderstadt and Hamilton, 1976).

3.1. Case 1: verification of CNN by conventional point kinetics (PK)

The PK model is the space independent model of neutron flux variation. CNN model could be reduced to one cell in space to be compared with PK model. Due to simplicity of PK model, it could be readily solved numerically by MATLAB functions. The results of PK model and CNN reduced geometry are illustrated in Figs. 5 and 6.

In this case, we compared the CNN results with numerical solution for a positive step reactivity insertion of 0.5 and 0.1 dollars. The aim is to investigate reactor power following a positive step change in reactivity, i.e., prompt jump. The PK model has been used to calculate the reactor power transient with an appropriate neutron mean generation time (Duderstadt and Hamilton, 1976). This transient is implemented via a change in boric acid concentration in the core. It is also replicated in CNN by means of a step change in resistors values in all cells of the CNN [Eq. (12)]. In comparing the results a percent error (PE) parameter is defined as follows:

$$\mathrm{PE} = \frac{\int_0^T \left(\frac{|\phi_{\mathrm{PK}}(t)-\phi_{\mathrm{CNN}}(t)|}{\phi\mathrm{PK}(t)}\right)\cdot \mathrm{d}t}{T} \times 100, \tag{20}$$

where T is the simulation time, $\phi_{\mathrm{PK}}(t)$ and $\phi_{\mathrm{CNN}}(t)$ are time varying neutron fluxes from PK and CNN models. The PE value was calculated to be 1.5% for this case study. The difference could be originated from numerical sources of error in both methods.

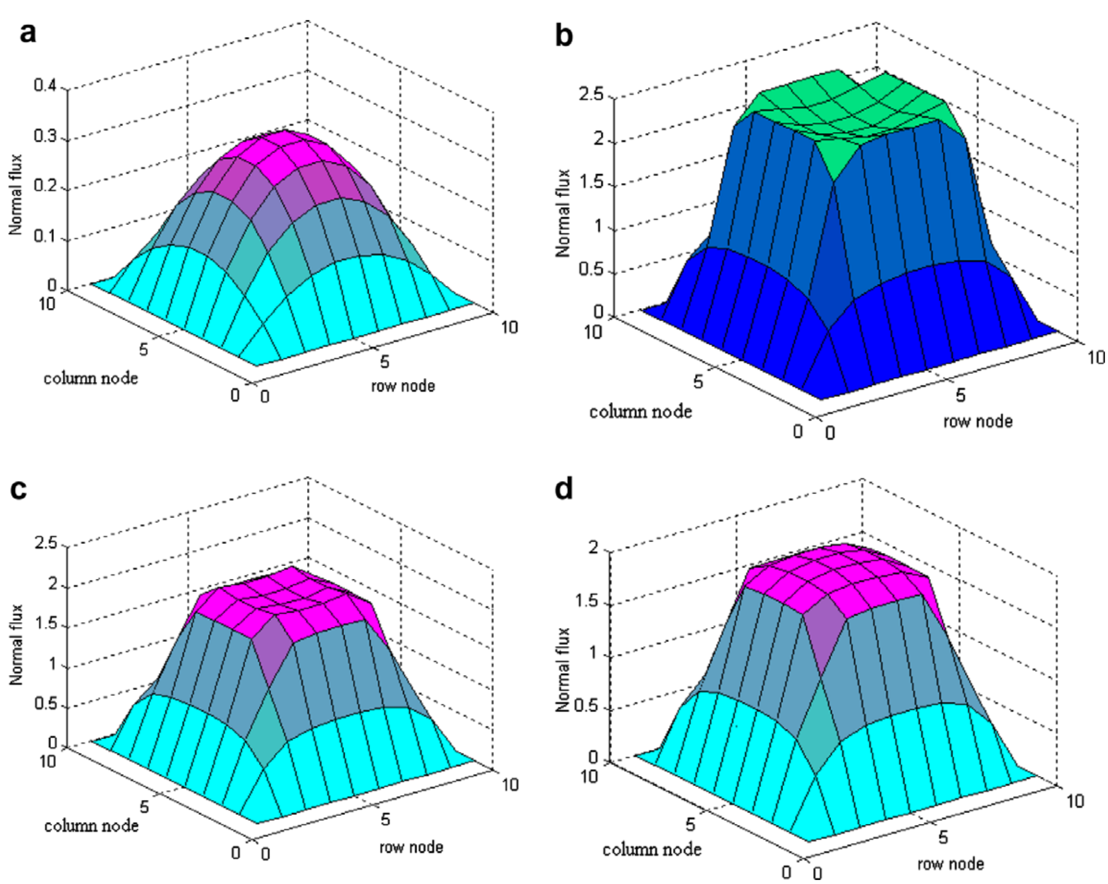

Fig. 7. Time variations of neutron flux at 14th node due to +0.5$ reactivity insertion. (a) Before perturbation, (b) $t = 160$ s, (c) $t = 250$ s and (d) $t = 300$ s.

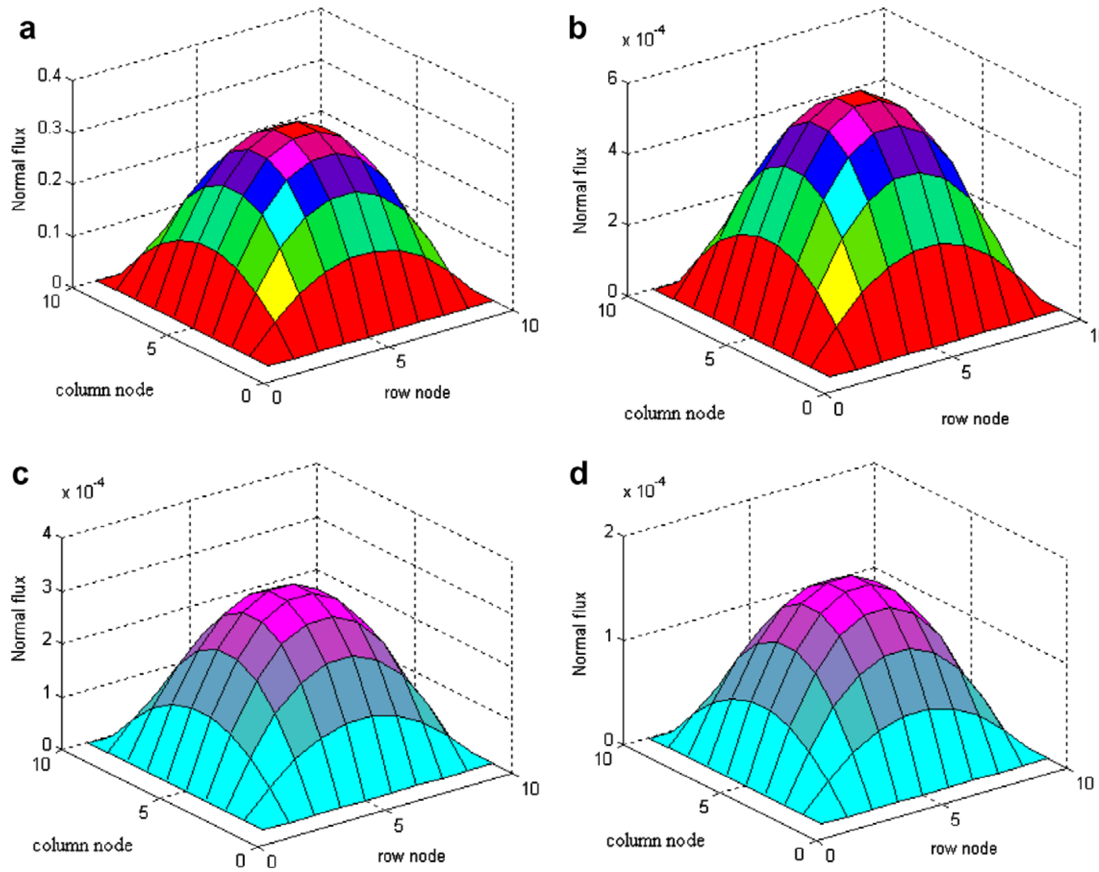

Fig. 8. Time variations of neutron flux at 14th node due to $-0.2\$$ reactivity insertion. (a) Before perturbation, (b) $t = 160$ s, (c) $t = 250$ s and (d) $t = 300$ s.

3.2. Case 2: neutron flux variation due to a local perturbation

A transient case involving control rod movements is studied for this case. The height of the core is divided in 18 nodes, To show the effect of a local perturbation, a positive reactivity of $+0.2\$$ is inserted into the central cell of the 14th node which is at $z = 85.5$ cm $\left(\frac{14}{18} \times 111\right)$. As can be seen in Fig. 7, the core flux increases rapidly at first, but as the feedback phenomenon takes control; the flux reduces with time slowly. Eventually, the flux magnitude is raised to a new value as the result of positive reactivity insertion.

The same scenario but with negative reactivity insertion of $-0.2\$$ at $z = 85.5$ cm is repeated and the results are shown in Fig. 8. With negative reactivity insertion, the flux is reduced rapidly at first and continues to decrease slowly with time.

3.3. Case 3: control rod cluster movements

We considered the movement of a control rod cluster versus time as shown in Fig. 9. A cluster of 4 control rods located at $(x = 3,\ y = 3)$, $(x = 3,\ y = 4)$, $(x = 4,\ y = 3)$ and $(x = 4,\ y = 4)$ are assumed to move into and out of the core to simulate reactivity insertion and withdrawal. The time varying thermal neutron flux variations are illustrated in Fig. 10. For a cubical rector core in absence of a perturbation, fast and thermal spatial fluxes have a cosine shape (Fig. 10a). As the control rods move into the core during $100 \leqslant t \leqslant 250$ s flux dipping is illustrated in Fig. 10b and c. During the rods withdraw $(250 < t \leqslant$

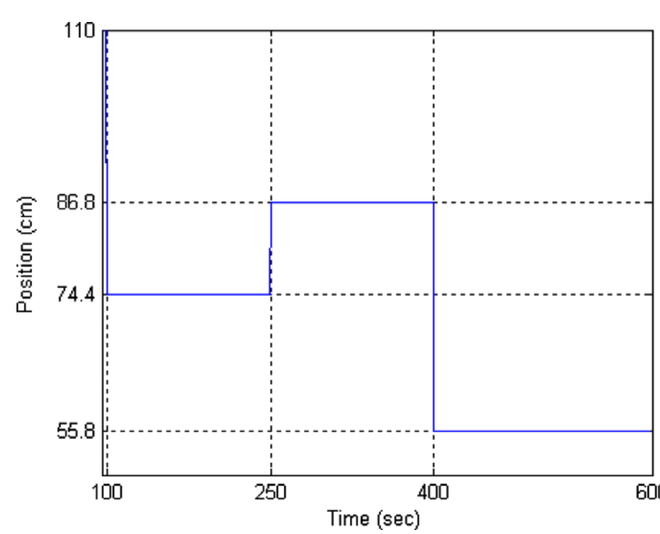

Fig. 9. Control rod clusters movement with time.

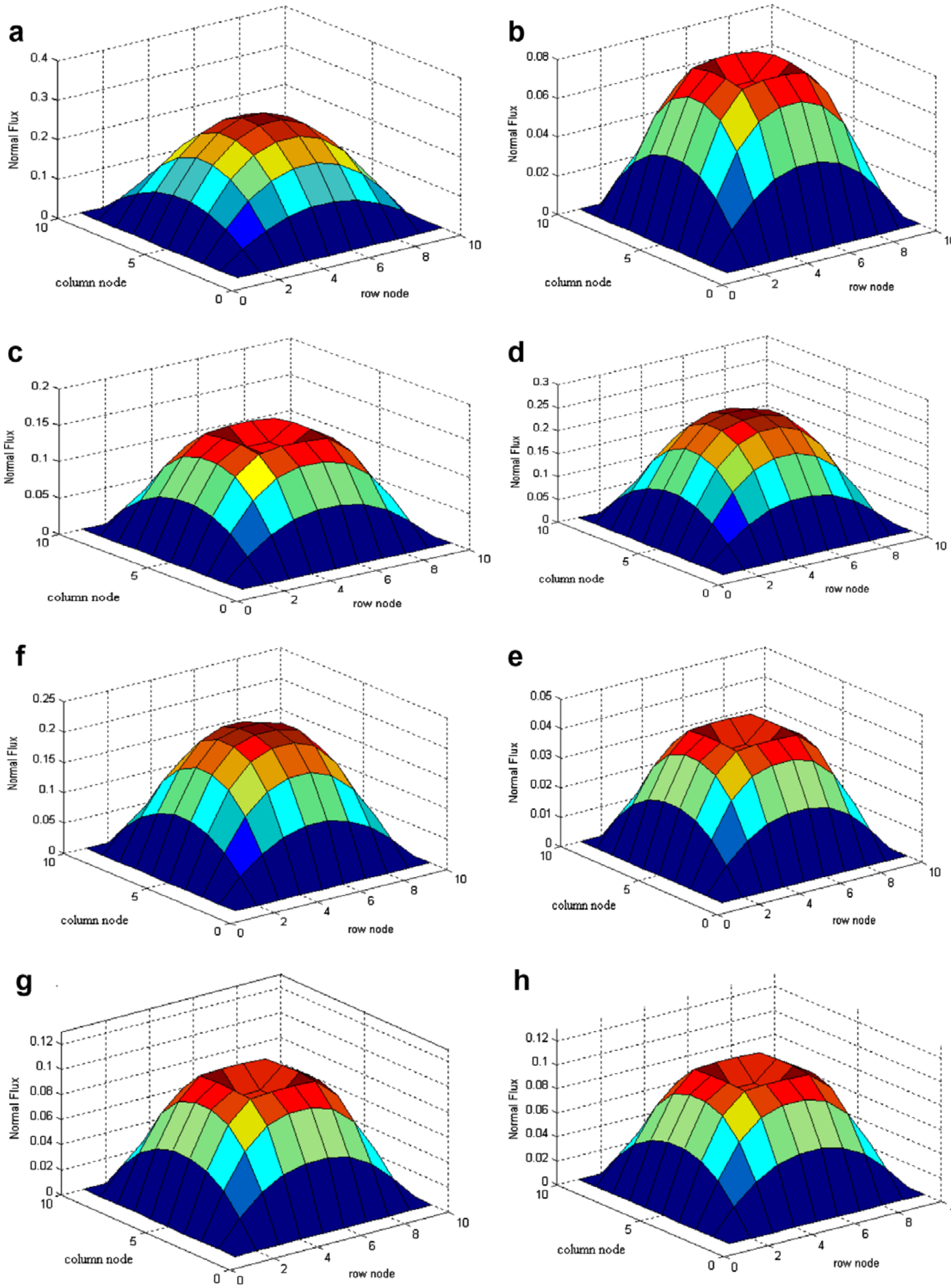

Fig. 10. Thermal neutron flux at 80.5 cm with movement of control rods (a) before movement, (b) $t = 105$ s, (c) $t = 145$ s, (d) $t = 260$ s, (e) $t = 390$ s, (f) $t = 410$ s, (g) $t = 490$ s and (h) $t = 600$ s.

400 s) thermal flux increases to a new equilibrium level, shown in Fig. 10d and e. The control rods move in during $400 < t \leqslant 600$ s and the core thermal flux is depressed to lower power as illustrated in Fig. 10f, g and h.

The reactor core axial flux is also an important parameter which its simulation results are shown in Fig. 11. During normal operation, the axial flux (thermal and fast) has a cosine shape and as the perturbation applies, flux shape depresses. The thermal neutron flux during rod movements at different time segments is plotted and compared against its normal operation. Hot spots development due to flux peaking may cause thermal stresses which may be further investigated in accident scenarios.

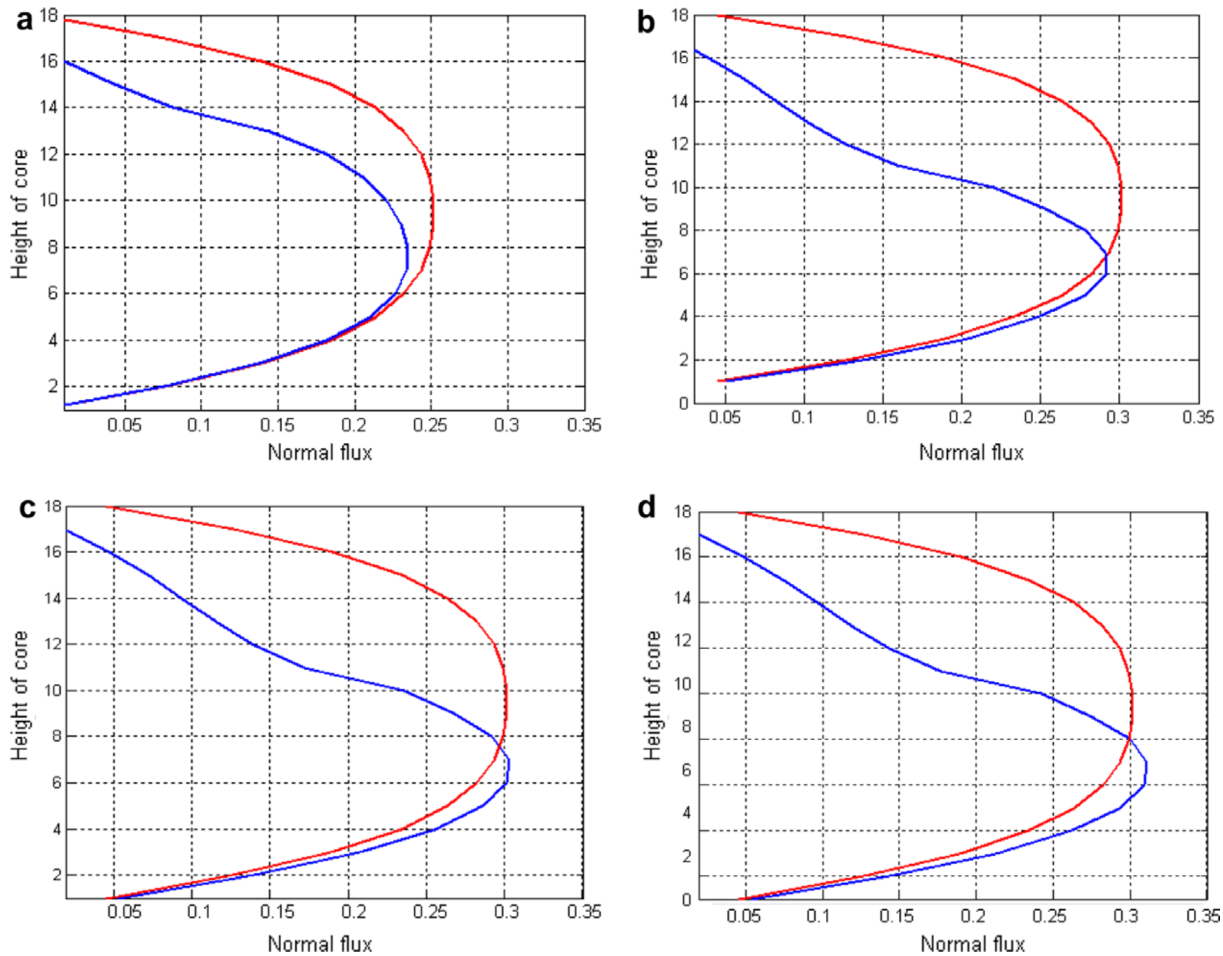

Fig. 11. Axial thermal flux variations with time during rods movement (a) $t = 160$ s, (b) $t = 450$ s, (c) $t = 550$ s and (d) $t = 600$ s.

4. Conclusion and suggestions

We have successfully simulated the dynamics of the primary loop of a typical nuclear reactor using a semi-analog medium, which is new for such simulations. This medium (HSpice) enables us to further develop VLSI integrated hardware, which could assist reactor operation and reactor training simulators. Computerized operator decision support is usually used online for the calculation of the time-space power distribution of an advanced reactor core (Lebrun, 1994). The stability analysis of the CNN model proves to be highly stable and relatively insensitive to non-physical perturbations. The CNN model was tested and verified using the classical point kinetic model. While classical methods of solution of such nonlinear PDEs were sensitive to numerical differentiation methods, the CNN could be readily adjusted for optimum performance.

The current practice for many simulators is based on the application of pre-calculated data generated by reliable thermal hydraulic codes. Our study could improve the current practices by providing online data readily produced by the CNN simulator.

Our model can be further improved by implementing the followings:

(1) Implementing cylindrical and spherical geometries for real core modeling.
(2) Improving the accuracy by increasing the spatial and energy resolutions.
(3) Implementing both fuel and moderator feedbacks with mass and energy balance for two phase flow to address accident scenarios in LWRs.
(4) Coupling with updated cross generation codes to improve the model's reliability and testing the model against reactor dynamic codes as well as experimental data.

References

Adelr, F.T., 1961. Reactor kinetics: integral equation formulation. J. Nucl. Energ., Parts A & B 15, 81.

Allerd, J.C., Carter, D.S., 1958. Kinetics of homogeneous power reactors of the LAPRE type. Nucl. Sci. Eng. 3, 482.

Balsi, A., Marongiu, Cimagalli, V., 1995. Electromagnetic field simulation using 3-D cellular neural networks. In: Proc. European Conf. Circuit Theory and Design (ECCTD'95), Istanbul, Turkey, pp. 987–990.

Blue, L.R., Hoffman, M., 1963. Generalized program for the numerical solution of space independent reactor kinetics equations, NAA – SR – Memo – 9197, North American Aviation.

Boroushaki, M., Ghofrani, M.B., Lucas, C., 2005. Simulation of nuclear reactor core kinetics using multilayer 3-D cellular neural networks. IEEE Trans. Nucl. Sci. 52 (3).

Brown, H.D., 1957. A general treatment of flux transients. Nucl. Sci. Eng. 12, 687.

Chua, L., Roska, T., 1993. The CNN paradigm. IEEE Trans. Circuits Syst. 40, 147–156.

Chua, L.O., Roska, T., 2002. Cellular neural networks and visual computing. Cambridge University Press, UK.

Chua, L.O., Hasler, M.G., Moschytz, S., Neirynck, J., 1995. Autonomous cellular neural networks: a unified paradigm for pattern formation and active wave propagation. IEEE Trans. Circuits Syst. I 42 (10), 559–577.

Civalleri, P.P., Gilli, M., 1995. Circuit models for linear and nonlinear waves. IEEE Trans. Circuits Syst. I 42 (10), 578–582.

Crounse, R., Chua, L.O., 1995. Methods for image processing and pattern formation in cellular neural networks: a tutorial. IEEE Trans. Circuits Syst. 42, 583–601.

Duderstadt, J.J., Hamilton, L.J., 1976. Nuclear Reactor Analysis. Wiley, New York, pp. 241–245.

Faghihi, F., Hadad, K., 2004. Numerical solutions of coupled differential equations and initial values using Maple software. Appl. Math. Comp. 155, 563–572.

Hadad, K., 2004. Numerical solution of multigroups point kinetic equations by simulink toolbox of Matlab software. In: Proc. 2nd Intl. Conf. Nucl. Sci. Tech.

Hansen, K.F., 1965. Stable numerical solutions of the reactor kinetics equations. Nucl. Sci. Eng. 22, 51.

Hetrick, D.L., 1993. Dynamics of Nuclear Reactors. American Nuclear Society, La Grange Park, IL.

Kaganove, J.J., 1960. Numerical solution of the one group space density independent reactor kinetics equations for neutron density given the excess reactivity. ANL – 6132, Argonne National Laboratory.

Keepin, G.R., Cox, C.W., 1960. General solution of the reactor kinetic equations. Nucl. Sci. Eng. 8, 670.

Kim, H., Son, H., Roska, T., Chua, L., 2005. High-performance Viterbi decoder with circularly connected 2-D CNN unilateral cell array. IEEE Trans. Circuits Syst. I 52 (10), 2208–2218.

Kozek, T., Chua, L.O., Roska, T.D., Wolf, R., Tetzaff, F., Puffer, F., Lotz, K., 1995. Simulation nonlinear waves and partial differential equations via CNN – part II: typical examples. IEEE Trans. Circuits Syst. 42 (10), 816–820.

Krstic, W., 2001. Cellular neural network to model and solve direct non-linear problem of steady-state heat transfer. IEEE Trans. Circuits Syst., 420–423.

Lebrun, J.P., 1994. Core control and protection principles for N4 plants, IAEA, Vienna, Italy, Consulting Meeting on Core Control and Protection Strategy of WWER-1000 Reactors Rep.

Porsching, T.A., 1966. Numerical solution of the reactor kinetics equations by approximate exponential. Nucl. Sci. Eng. 25, 183.

Ravetto, P., 1997. Reactivity oscillations in a point reactor. Ann. Nucl. Energ. 24 (4), 303–314.

Roska, T., Chua, L., Wolf, D., Kozek, T., Tetzlaff, R., Puffer, F., 1995. Simulating nonlinear waves and partial differential equations via CNN – part I and II. IEEE Trans. Circuits Syst. I 42 (10), 807–820.

Sanchez, J., 1989. On the numerical solution of the point reactor kinetics equations by generalized Runge–Kutta methods. Nucl. Sci. Eng. 103, 94–99.

Stacy, W.M., 2001. Nuclear Reactor Physics. Wiley, New York, pp. 599–665.

Vigil, J.C., 1967. Solution of the reactor kinetics equations by analytic continuation. Nucl. Sci. Eng. 29, 392–401.